习题 1.8

$$\int_{\ell} \mathrm{Re}\, z\, dz = \int_0^1 t\,(1+2it)\,dt = \frac{1}{2} + i$$

习题 1.9

(1) $$\frac{\partial^n \psi}{\partial t^n}\Big|_{t=0} = \frac{n!}{2\pi i}\oint_L \frac{\psi(\xi, x)}{(\xi-0)^{n+1}}\,d\xi = \frac{n!}{2\pi i}\oint_L \frac{e^{2\xi x-\xi^2}}{\xi^{n+1}}\,d\xi \quad (z=0 \text{ 在 } L \text{ 内})$$

(2) $$\frac{\partial^n \psi}{\partial t^n}\Big|_{t=0} = \frac{n!}{2\pi i}\oint_L \frac{e^{x^2-(x-\xi)^2}}{\xi^{n+1}}\,d\xi = e^{x^2}\,\frac{n!}{2\pi i}\oint_{L'} \frac{e^{-z^2}}{(x-z)^{n+1}}\,(-dz)$$

$$= (-1)^n e^{x^2}\,\frac{n!}{2\pi i}\oint_{L'} \frac{e^{-z^2}}{(z-x)^{n+1}}\,dz = (-1)^n e^{x^2}\frac{d^n e^{-x^2}}{dx^n} \quad ($$

习题 1.10

(1) $$\oint_{|z|=1} \frac{dz}{\cos z}$$

解. 在 $|z|<1$ 内，$\frac{1}{\cos z}$ 没有奇点，所以 $\oint_{|z|=1}\frac{dz}{\cos z} = 0$.

(2) $$\oint_{|z|=1} \frac{dz}{z^2+2z+4}$$

解. 若 $z^2+2z+4=0$，则 $z=-1\pm\sqrt{3}i \Rightarrow |z| = 2 > 1$

所以 $\frac{1}{z^2+2z+4}$ 在 $|z|<1$ 内没有奇点 $\Rightarrow \oint_{|z|=1}\frac{1}{z^2+2z+4}\,dz = 0$

(3) $\oint_{|z|=2} z^3\cos z\, dz = 0$ （根据柯西积分定理）

(4) $$\oint_{|z|=4}\left(\frac{4}{z+1}+\frac{1}{z+2i}\right)dz$$

解. 在 $|z|<4$ 内，$\frac{4}{z+1}$ 有奇点 $z=-1$，$\frac{1}{z+2i}$ 有奇点 $z=-2i$

$$\Rightarrow \oint_{|z|=4}\left(\frac{4}{z+1}+\frac{1}{z+2i}\right)dz = \oint_{|z+1|=0.5}\frac{4}{z+1}\,dz + \oint_{|z+2i|=0.5}\frac{1}{z+2i}\,dz$$

$$= 2\pi i\cdot(4+1) = 10\pi i$$

(5) $\oint_{|z|=2} \frac{\sin z}{(z-\frac{\pi}{2})^2} dz$

解. 原式 $= \frac{2\pi i}{1!} \frac{d}{dz}(\sin z)\Big|_{z=\frac{\pi}{2}} = 0$.

习题 1.11

計算 $\oint_{|z|=1} \frac{dz}{z+2} = 0$, 同时 $\oint_{|z|=1} \frac{dz}{z+2} = \int_0^{2\pi} \frac{e^{i\theta}\cdot i}{2+\cos\theta+i\sin\theta} d\theta$

$$= \int_0^{2\pi} \frac{1}{5+4\cos\theta}\cdot(-\sin\theta+i\cos\theta)(2+\cos\theta-i\sin\theta)\,d\theta$$

$$= \int_0^{2\pi} \frac{i(2\cos\theta+1)+(\cos\theta\sin\theta-2\sin\theta-\cos\theta\sin\theta)}{5+4\cos\theta}\,d\theta$$

$$\Longrightarrow \int_0^{2\pi} \frac{2\cos\theta+1}{5+4\cos\theta}\,d\theta = 0$$

~~又知 $\mathrm{Im}\left(\int_{|z|=1,\ \mathrm{Im}(z)>0} \frac{1}{z+2} dz\right) = \int_0^{\pi} \frac{1+2\cos\theta}{5+4\cos\theta} d\theta$~~ =

又知 $\mathrm{Im}\left(\int_l \frac{1}{z+2} dz\right) = \int_0^{\pi} \frac{1+2\cos\theta}{5+4\cos\theta} d\theta = \mathrm{Im}\left(\ln\frac{-1+2}{1+2}\right) = 0$

习题 1.12

(1) $R = \lim\limits_{k\to\infty} \frac{\frac{1}{k}}{\frac{1}{k+1}} = 1$, 收敛圆 $|z-i|=1$

(2) $R = \lim\limits_{k\to\infty} \frac{k^{\ln k}}{(k+1)^{\ln(k+1)}} = \lim\limits_{k\to\infty} e^{(\ln k)^2-(\ln(k+1))^2} = \lim\limits_{k\to\infty} e^{\ln\left(\frac{k}{k+1}\right)\ln(k(k+1))} = 1$

收敛圆 $|z-2|=1$

(3) $R = \lim\limits_{k\to\infty} \frac{k!}{(k+1)!}\cdot\frac{(k+1)^{k+1}}{k^k} = \lim\limits_{k\to\infty}\left(1+\frac{1}{k}\right)^k = e$, 收敛圆 $|z|=e$.

(4) $R = \lim\limits_{k\to\infty} \frac{1}{(k^k)^{\frac{1}{k}}} = 0$, 收敛圆 $z=3$

习题 1.13

(1) $f(z) = \sqrt[3]{z} = z^{\frac{1}{3}} = (z-i+i)^{\frac{1}{3}} = i^{\frac{1}{3}}\left(1+\frac{z-i}{i}\right)^{\frac{1}{3}} = i^{\frac{1}{3}}\sum_{n=0}^{\infty}\binom{\frac{1}{3}}{n}\left(\frac{z-i}{i}\right)^n = \sum_{n=0}^{\infty}\binom{\frac{1}{3}}{n} i^{\frac{1}{3}-n}\cdot(z-i)^n$.

$$= \sum_{n=0}^{\infty} \frac{\prod_{k=0}^{n-1}\left(\frac{1}{3}-k\right)}{n!}\cdot i^{\frac{1}{3}-n}(z-i)^n \quad \boxed{定义\ \binom{\frac{1}{3}}{0}=1}$$

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(2) $f(z) = \ln(1+e^z)$

$$f'(z) = \frac{e^z}{1+e^z} = 1 - \frac{1}{1+e^z} = 1-(1-e^z+e^{2z}-e^{3z}+\cdots) = \sum_{k=1}^{\infty}(-1)^{k-1}e^{kz}$$

$$= \sum_{k=1}^{\infty}\left[(-1)^{k-1}\sum_{n=0}^{\infty}\frac{(kz)^n}{n!}\right] = \sum_{n=0}^{\infty}\sum_{k=1}^{\infty}(-1)^{k-1}\cdot\frac{(kz)^n}{n!} = \sum_{n=0}^{\infty}\left(\sum_{k=1}^{\infty}\frac{(-1)^{k-1}k^n}{n!}\right)z^n$$

$$f(z) = \int\sum_{n=0}^{\infty}\left(\sum_{k=1}^{\infty}\frac{(-1)^{k-1}k^n}{n!}\right)z^n\,dz = \sum_{n=0}^{\infty}\left(\sum_{k=1}^{\infty}\frac{(-1)^{k-1}k^n}{(n+1)!}\right)z^{n+1} = \sum_{n=1}^{\infty}\left(\sum_{k=1}^{\infty}\frac{(-1)^{k-1}k^{n-1}}{n!}\right)z^n$$

(3) $f(z) = \int_0^z \frac{\sin\xi}{\xi}\,d\xi$, $f'(z) = \frac{\sin z}{z} = \dfrac{\sum_{k=0}^{\infty}\frac{(-1)^k z^{2k+1}}{(2k+1)!}}{z} = \sum_{k=0}^{\infty}\frac{(-1)^k}{(2k+1)!}z^{2k}$

$$f(z) = \sum_{k=0}^{\infty}\frac{(-1)^k}{(2k+1)(2k+1)!}z^{2k+1}$$

(4) $\cancel{\frac{1}{z^2} = \frac{1}{(z+1-1)^2}}$ $f(z) = \frac{1}{z^2}$, $\frac{1}{z} = \frac{1}{-1+(z+1)} = -\left(\sum_{n=0}^{\infty}(z+1)^n\right)$

$$f(z) = \frac{d}{dz}\left(-\frac{1}{z}\right) = \sum_{n=0}^{\infty}n(z+1)^{n-1} = \sum_{n=1}^{\infty}n(z+1)^{n-1} = \sum_{n=0}^{\infty}(n+1)(z+1)^n$$

(5) $$\sin^2 z = \frac{1-\cos 2z}{2} = \frac{1}{2}\left(1-\sum_{n=0}^{\infty}\frac{(-1)^n(2z)^{2n}}{(2n)!}\right) = \frac{1}{2}\sum_{n=1}^{\infty}\frac{(-1)^{n+1}(2z)^{2n}}{(2n)!} = \sum_{n=1}^{\infty}\frac{(-1)^{n+1}2^{2n-1}}{(2n)!}z^{2n}$$

(6) $$\frac{2}{z^2-4z+3} = \frac{2}{-1+(z-2)^2} = -\left(\frac{2}{1-(z-2)^2}\right) = -2\sum_{n=0}^{\infty}(z-2)^{2n} = \sum_{n=0}^{\infty}(-2)(z-2)^{2n}$$